Maths (Introduction to Trigonometry) (Ch-8)

Q1) For $0° \le \theta \le 90°$, the maximum value of $\frac{1}{\sec\theta}$

(a) 1 (b) 0 (c) undefined (d) $\frac{\sqrt{3}}{2}$

Q2) If $4\tan\beta = 3$, then $\frac{4\sin\beta - 3\cos\beta}{4\sin\beta + 3\cos\beta} =$

(a) 0 (b) $1/3$ (c) $2/3$ (d) $7/25$

Q3 Given that $\sin\theta = \frac{a}{b}$, then $\tan\theta$ is equal to

(a) 0 (b) $1/3$ (c) $2/3$ (d) $7/25$

(a) $\frac{b}{\sqrt{a^2+b^2}}$ (b) $\frac{b}{\sqrt{b^2-a^2}}$ (c) $\frac{a}{\sqrt{a^2-b^2}}$ (d) $\frac{a}{\sqrt{b^2-a^2}}$

Q4) If $a\cos\theta - b\sin\theta = c$, then $a\sin\theta + b\cos\theta = ?$

Q5) If $\sec\theta - \tan\theta = m$, then value of $\sec\theta + \tan\theta$ is

Q6 If $\sin\alpha = \frac{\sqrt{3}}{b}$ Assertion: If $x = a\cos\theta$ and $y = b\sin\theta$, then $b^2x^2 + a^2y^2 = a^2b^2$

Reason: $\cos^2\theta + \sin^2\theta = 1$

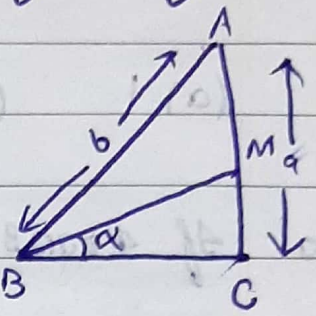
Q7 If $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ & $z = r\cos\theta$, then

(a) $x^2+y^2+z^2 = r^2$ (c) $x^2+y^2-z^2 = r^2$

(b) $x^2-y^2+z^2 = r^2$ (d) $z^2+y^2-x^2 = r^2$

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Q8) In given figure, $AM = AC$ & $\angle C$ is a right angle, then $\sin^2\alpha - \cos^2\alpha$ is equal to



(a) $\frac{4b^2-3a^2}{5a^2-4b^2}$ (b) $\frac{5a^2-4b^2}{4b^2-3a^2}$ (c) $\frac{9a^2-5b^2}{3b^2-4a^2}$

(d) $\frac{3b^2-4a^2}{4a^2-5b^2}$

Q9) If A & B are complementary angles, then

(a) $\sin A = \sin B$ (b) $\cos A = \cos B$ (c) $\tan A = \tan B$ (d) $\sec A = \operatorname{cosec} B$

Q10) $\frac{\cos(90^\circ-\theta)\sec(90^\circ-\theta)\tan\theta}{\cos(90^\circ-\theta)\sin(90^\circ-\theta)} + \frac{\tan(90^\circ-\theta)}{\cot\theta}$

is

(a) 1 (b) -1 (c) 2 (d) -2

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Maths Revision

Q1- A 30 m high building has a car standing on the road in front of it. The angle of depression of the car from the top of the building is 30°.

i) What is the angle of elevation of the top of the building from the car?

ii) Which trigonometric ratio will be used to find the distance from the building, of the car?

iii) Find the distance of the car from the foot of the building.

Q2- A boy flies a kite. The kite is at a height of 40 m above the ground. The string makes an angle of 60° with the horizontal.

i) Find the length of the string.

ii) Find the horizontal distance of the kite from the boy.

iii) Which trigonometric ratio relates the height of the kite to the length of the string?

Q3- A person observes the top of a 20 m high tower. From point A, the angle of elevation is 30°. He then moves towards the tower to point B, where the angle becomes 60°.

i) Find the distance of point A from the tower.

ii) Find the distance of point B from the tower.

iii) Find the distance travelled from A to B.

Q4- A vertical pole casts a shadow of length $10\sqrt{3}$ m where the angle of elevation of the sun is 30°.

i) Which trigonometric ratio can be used to calculate the height of the pole?

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ii) Find the height of the pole.

iii) If the angle of elevation of the sun becomes 45°, will the shadow become longer or shorter?

Q5- A lighthouse is 60 m high. A boat is observed from its top at an angle of depression of 45°.

i) What is the angle of elevation of the lighthouse from the boat?

ii) Find the distance of the boat from the foot of the lighthouse.

iii) If the boat moves 20 m far away, will the angle of elevation increase or decrease?

Q6- An aeroplane is flying at a height of 1000 m above the ground. From a point on the ground, its angle of elevation is 30°.

i) Find the horizontal distance of the aeroplane from the observer.

ii) Which trigonometric ratio can be used to find the horizontal distance of the aeroplane from the observer?

iii) Find the horizontal distance when the angle of elevation is 45°.

Q7- From the top of a 50 m high building, the angles of depression of two points A and B on the same side of the building are 30° and 60° respectively.

i) Which point is farther from the building?

ii) Find the distance of point A from the foot of the building.

iii) Find the distance of point B from the foot of the building.

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iv) Find the distance between A and B.

Q8- A ladder of length 20 m is placed against a vertical wall. It makes an angle of 60° with the ground.

i) Identify the hypotenuse of the right triangle.
ii) Find the height reached by the ladder on the wall.
iii) If the angle becomes 30°, what will be the height reached by the ladder?

Q9- From the top of a 60 m high tower, the angles of depression of two cars A and B, situated on the same side of the tower are 30° and 45° respectively.

i) Which car is nearer to the tower?
ii) Find the distance of car A from the foot of the tower.
iii) Find the distance of car B from the foot of the tower.
iv) Find the distance between the two cars.

Q10- A 10 m tall pole is placed on top of a building. From the point on the ground, the angle of elevation of the top of the building is 30°, while that of the top of the pole is 45°. The observer is $20\sqrt{3}$ m from the building.

i) Find the height of the building.
ii) Find the total height of the building and pole.
iii) If the observer moves closer, what happens to the angles of elevation.

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Maths (TRIANGLES)

Q.1 In ΔABC, a line DE is parallel to the base BC such that it cuts side AB at D and side AC at E. If the ration of AD : DB is 2 : 3 and the total length of AC is 15 cm, find the exact length of AE.

Q.2 In ΔABC, two points D and E are taken on side AB such that AD = BE. A line passing through D parallel to BC intersects AC at P, and another line passing through E parallel to BC insects AC at Q. Prove that AP = QC.

Q.3 In a trapezium ABCD where AB || DC, the two diagonals intersect each other exactly at point O. If AO = 3x − 1, BO = 2x + 1, CO = 5x − 3 and DO = 6x − 5, find the value of x.

Q.4 If $\Delta ABC \sim \Delta DEF$, and we draw a median AM in ΔABC and a median DN in ΔDEF, prove that the ration of the medians (AM/DN) is equal to the ratio of their corresponding side (AB/DE).

Application Based Question

Q.5 A girl of height 90 cm is walking away from the base of a lamp-post at speed of 1.2 m/s. If the lamp is 3.6 m above the ground, how long will her shadow be on the ground after exactly 4 seconds of walking?

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Maths [Polynomials]

1) If $ax^2 + bx + c$ has no real zeros & $a+b+c<0$, then

a) $c = 0$
b) $c > 0$
c) $c < 0$
d) cant be defined

2) Zeros of $7y^2 - \frac{11}{3}y - \frac{32}{3}$ are:

3) If the graph of ax^2+bx+c opens downward, then

a) $a > 0$
b) $a < 0$
c) $a \geq 0$
d) cant be defined

4) The coordinate of the vertex of a parabola is

a) $(-b, c)$
b) (a, b)
c) $\left(\frac{-b}{a}, \frac{D}{4a}\right)$
d)

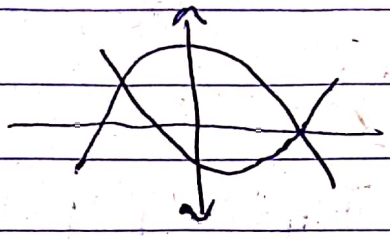
5) The zeros of $x^2 - 2x + 1$ are α, β. Form a polynomial of zeros $(\alpha - 1)$ and $(\beta - 1)$

6) Zeros of $(k^2+4)x^2 + 7x + 4k$ are reciprocal of each other. $k = ?$

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7) $3x^2 + 6x + k$ has zeros α, β and $\alpha + \beta + \alpha\beta = -2/3$, $k = ?$

8) graph of 2 poly , The no. of common zeros?



9) zeros of $x^2 + \frac{x}{0} - 2$ are ?

10) $x^2 - ax - b$, $\alpha^2 + \beta^2 = ?$

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